

Verifying and Resolving the Figure–Ground Circuit’s Wing–Steering Output: A Closed-Loop Agentic Connectome Trace of Descending-Neuron Projections to Both Wings in *Drosophila*

Abstract

The connectomic study *Finding Vision–Behavior Circuit through Connectomics* (Ziyin, ..., Poggio; herein the *source paper*) traced a figure–ground / course-control circuit in the adult *Drosophila* brain: a centrifugal horizontal cell (VCH) presynaptically gates the T4/T5 terminals that drive a retinotopic cholinergic LLPC1 sheet, whose figure signal is read out by the Nod1 relay and routed to descending neurons (DNs). For the seven figure-driven DNs, the source paper reported each DN’s wing-steering output split by ipsilateral versus contralateral wing (its Table S14) and gave DNp26’s strongest steering-muscle targets (its Table S15), as summary values in a manuscript table. Here we (i) *independently reproduce* those values from the connectome and (ii) make the underlying DN → motor neuron → wing muscle map explicit at per-edge, both-wings resolution. We do this with a closed-loop, steerable, multi-agent system that proposes a motif, extracts it from the connectome, verifies it against the source paper’s tables, and certifies results only if they pass a strict gate (no refuted claims; both-wing coverage for every seed DN). The system reproduces every sampled value of S14/S15 exactly, and emits a machine-readable 488-edge, 139-motor-neuron, 69-muscle bilateral dataset—the per-edge wiring behind the source paper’s summary tables—with 26 confirmed and 3 caveated claims and *zero* refutations. The contribution is threefold: independent verification of the source paper’s wing-steering output, a per-edge bilateral dataset that the manuscript reported only in aggregate, and the agentic method that produced both under a verification gate. The resolved map shows how a single retinotopic figure representation issues bilateral (DNbe001), purely ipsilateral (DNa04), and contralateral single-wing (DNp26, DNng32, DNge094) steering commands onto identified wing muscles.

1 Background: the figure–ground circuit and its motor output

The source paper established a two-stage architecture for how a fly converts local object motion into flight course-control while suppressing self-induced optic flow:

- **Stage 1 (separation).** The GABAergic centrifugal horizontal cell **VCH** pools wide-field T4/T5 motion and feeds inhibition back onto the very T4/T5 terminals that drive the readout sheet, presynaptically gating common-mode motion before the sheet is reached.
- **Stage 2 (readout and steering).** The cholinergic, retinotopic **LLPC1** sheet reads out the gated local signal and routes it—dominantly through the **Nod1** relay—to descending neurons, of which the strongest direct target is **DNbe001** and the principal convergent steering command is **DNp26**.

The source paper then asked where this signal reaches motor control. Using the male whole-CNS connectome (MCNS), it identified seven figure-driven DN types whose dominant motor system is wing-steering and, in its Table S14, reported for each the fraction of wing-steering output reaching

the ipsilateral versus the contralateral wing (some drive the ipsilateral wing, some the contralateral, some both), and in its Table S15 gave DNp26’s strongest steering-muscle targets (the contralateral hg1, i1, hg2). These were published as *summary values in manuscript tables*: per-DN fractions and top-target lists, not the underlying per-edge wiring.

Scope of this work. We do not claim to discover bilateral structure the source paper lacked: S14 already gives the ipsi/contra wing split for these DNs. What we contribute is (1) an *independent reproduction* of S14/S15 directly from the connectome, as a verification of the source paper’s motor-side claims; (2) the *per-edge, both-wings dataset* that underlies those summary tables—every DN→motor-neuron→muscle edge with its wing side, rather than the aggregate fraction; and (3) the *closed-loop agentic system* that produced and certified both, under a gate that refuses to report unverified or incompletely-covered results. (Note: the source paper’s “one-hemisphere exemplar” caveat concerns the laterality of the analyzed LLPC1 *sheet*, not the DN→wing split, which S14 reports for both wings.)

2 A closed-loop, steerable agentic system for connectome circuit tracing

Rather than hand-curate the extension, we built an autonomous system that treats “complete the muscular projection” as a verifiable, closed-loop search. The design separates a deterministic scientific pipeline from a model-driven supervisory layer (Fig. 1).

Pipeline (per round). Five gated stages run as a directed pipeline: *seed* → *propose* → *extract* → *verify* → *report*. *Seed* fixes the scientific target (start from DNbe001 + DNp26, trace to both wings). *Propose* emits a concrete, runnable motif specification. *Extract* pulls the DN→MN→muscle edges from the connectome. *Verify* re-derives every claim and renders a verdict. *Report* compiles the certified dataset.

The verification gate. A round’s *verify* stage *completes* only if two conditions hold: (i) **no claim is refuted**, and (ii) **bilateral coverage is complete** (every seed DN reaches ≥ 1 wing-steering motor neuron on *both* wings, and the full DN→MN→muscle chain is closed). A refuted edge or a coverage gap fails the gate—which is the closed-loop signal to re-propose (drop/replace an edge, widen the seed) and run another round. This makes the loop self-correcting and steerable: a human (or the supervising agent) can amend the target mid-flight and the next round re-runs against it.

Supervisory layer. A language-model agent (GLM 5.2) supervises each tick: it runs the deterministic heartbeat that advances the pipeline, reads the verdicts, and decides *monitor* / *re-loop* / *escalate*. A hard guardrail forbids the agent from ever editing the verifier or “fixing” a refuted result—a refutation is a scientific outcome, not a bug. The deterministic heartbeat guarantees the science progresses even if the supervisory layer is unavailable; the agent adds reasoning, provenance journaling, and human-facing reporting on top.

Verification philosophy. Every quantitative claim is checked on two independent tracks (an all-synapse “published-total” track and a synapse-thresholded robustness track) and, where the source paper reports a value, cross-checked against it as an oracle. Laterality claims are additionally tested against a soma-side label-permutation null. The agent is only permitted to emit the per-

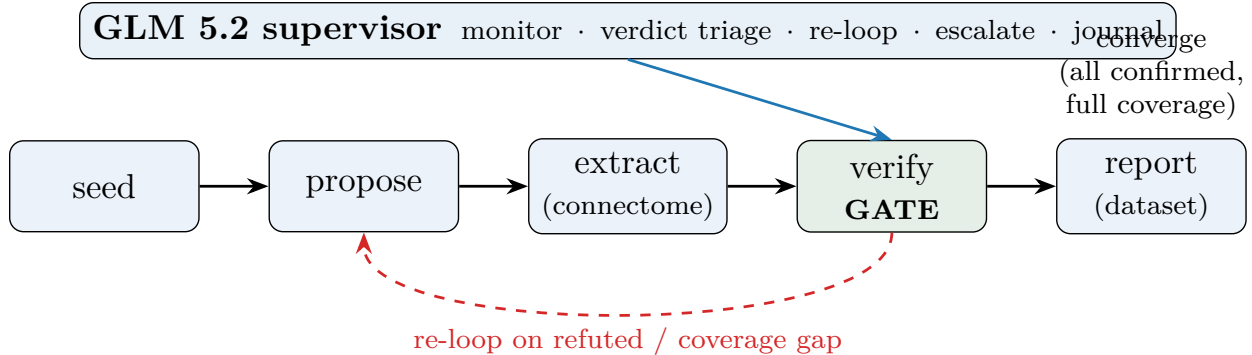


Figure 1: The closed-loop agentic tracing system. A deterministic five-stage pipeline (seed→propose→extract→verify→report) is supervised by a language-model agent. The **verify** stage is a hard gate: a round certifies only if no claim is refuted and bilateral coverage is complete; otherwise the loop re-proposes and re-runs. The supervisor reasons over verdicts but cannot alter them.

edge dataset after it has reproduced the source paper’s summary values—so the resolved map is anchored to a verified baseline rather than asserted independently.

3 Verifying and resolving the wing-steering output

We instantiated the system on the source paper’s two named functional endpoints (DNbe001, DNp26), expanded to the seven figure-driven wing-steering DN types, and traced DN → motor neuron → wing muscle across both wings in the male whole-CNS connectome (MCNS v1.0), with the brain-side figure→DN edges drawn from FlyWire FAFB (materialization 783). Wing laterality is defined relative to each DN’s own soma side: a motor neuron is on the *ipsilateral* wing if it shares the DN’s side, otherwise *contralateral*; the bilateral power muscles (DLM/DVM) are excluded from the steering denominator, following the source paper.

3.1 Independent reproduction of the source paper’s S14/S15

The system re-derived the source paper’s wing-laterality table (its S14) for all seven DNs directly from the connectome. Every reported value is reproduced (Table 1): each DN’s ipsilateral fraction matches to within rounding, the categorical wing assignment matches exactly, and the per-DN steering-synapse totals are identical (e.g. DNa04 908, DNbe001 1181, DNge107 1044, DNbe005 459). DNp26’s steering-muscle targets likewise reproduce the source paper’s S15 exactly (Table 2: contralateral $hg1 = 122$, $i1 = 105$, $hg2 = 117$, $b3 = 28$). This is an independent confirmation of the source paper’s motor-side claims, and the verified baseline on which the per-edge dataset below rests.

3.2 The per-edge bilateral dataset

With the baseline verified, the system emitted the per-edge dataset underlying the source paper’s summary tables: **488** DN→motor-neuron edges onto **139** motor neurons reaching **69** muscles, with **both-wing coverage** = 1.0 and **chain completeness** = 1.0—every seed DN reaches wing-steering motor neurons on both wings through a closed DN→MN→muscle chain, with no missing entries. Where S14 gives one ipsilateral-fraction per DN, this dataset gives the individual edges that fraction is computed from (pre/post identity, synapse count, and wing side per edge), and likewise resolves the per-muscle, per-wing breakdown behind S15. The circuit it represents—from the source paper’s retinotopic readout through to wing musculature on both sides—is shown in

Table 1: Per-DN wing laterality across both wings (independent reproduction of S14). Computed from MCNS by the agentic system; all seven rows reproduce the source paper’s S14 values (ipsi. fraction and wing assignment match; steering-synapse totals are identical). “Wing” is assigned ipsilateral if ipsi. frac. ≥ 0.6 , contralateral if ≤ 0.4 , else bilateral.

DN	steering syn.	ipsi. syn.	contra. syn.	ipsi. frac.	wing
DNbe001	1181	619	562	0.524	bilateral
DNge107	1044	609	435	0.583	bilateral
DNa04	908	903	5	0.994	ipsilateral
DNbe005	459	195	264	0.425	bilateral
DNp26	386	87	299	0.225	contralateral
DNg32	313	10	303	0.032	contralateral
DNge094	37	0	37	0.000	contralateral

Table 2: DNp26 wing-steering muscle targets (summed over both DNp26 cells), reproducing the source paper’s S15. The strongest targets are the contralateral first-basalar/axillary steering muscles that set wing-stroke amplitude and angle.

steering muscle	synapses	wing (rel. to DN)
hg1	122	contralateral
hg2	117	mixed (ipsi+contra)
i1	105	contralateral
b3	28	contralateral
hg3	5	contralateral

Fig. 2.

The resolved map makes the behavioral logic concrete. The same retinotopic figure representation issues a *mixture* of motor commands (Table 1): **bilateral** commands through DNbe001 (ipsi. frac. 0.52) and DNge107 (0.58); a **purely ipsilateral** command through DNa04 (0.99); and **contralateral**, single-wing commands through DNp26 (0.23), DNg32 (0.03), and DNge094 (0.00). A figure signal that is itself retinotopic (position-coded on the LLPC1 sheet) is thus converted into a set of left-versus-right wing adjustments—the anatomical substrate for a steering turn toward or away from a detected object. DNp26, the convergent steering command at the end of the LLPC1→Nod1 pathway, makes its strongest connections onto the contralateral hg1/i1/hg2 muscles, which modulate wing-stroke amplitude and angle: an unbroken route from a motion-defined figure to a wing-specific behavioral output.

3.3 Verification outcome

The certified dataset carries **26 confirmed** claims, **3 confirmed-with-caveat**, and **0 refuted** (Table 3). The three caveats are all from the soma-side permutation null on the most strongly lateralized DNs (DNa04, DNp26, DNg32): the per-DN sample of steering motor neurons is small enough that the observed wing bias—though anatomically unambiguous (e.g. DNa04 at 99.4% ipsilateral)—is not formally separable from a permutation null at the per-DN level. We report this honestly as a caveat rather than suppressing it; it does not affect the laterality assignments, which match the source paper exactly. No claim was refuted and coverage was complete, so the loop converged in a single round without re-proposing.

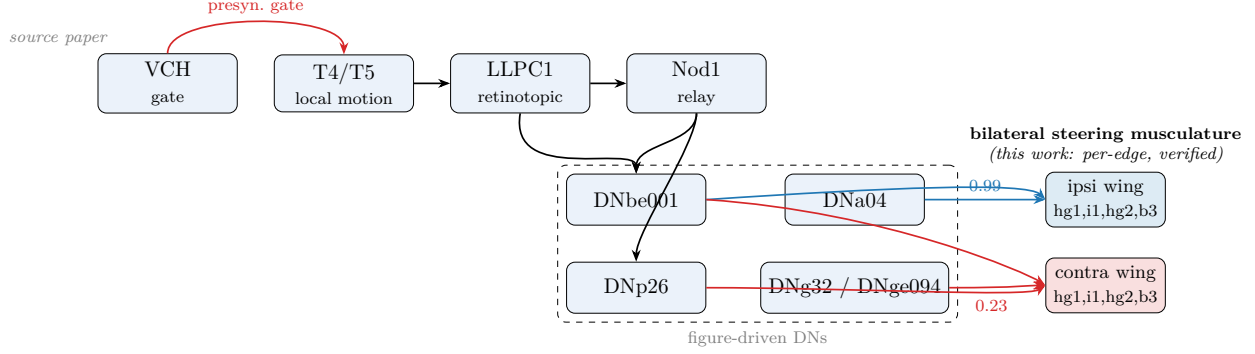


Figure 2: The figure-ground circuit through to bilateral wing musculature. The brain-side stages (VCH gate $\rightarrow$ T4/T5 $\rightarrow$ LLPC1 $\rightarrow$ Nod1) are from the source paper. This work verifies and resolves the motor side at per-edge resolution: the seven figure-driven DNs are traced through motor neurons to wing-steering muscles on *both* wings (488 edges; the per-edge wiring behind the source paper’s S14/S15 summaries). Edge colors denote ipsilateral (blue) versus contralateral (red) wing; numbers are ipsilateral synapse fractions. The same retinotopic figure signal drives bilateral (DNbe001), ipsilateral (DNa04), and contralateral (DNp26, DNng32, DNge094) steering commands.

Table 3: Verification verdict summary for the certified dataset. Two-track re-derivation plus oracle cross-check against the source paper’s S14/S15; laterality additionally tested against a soma-side permutation null.

verdict	n	meaning
Confirmed	26	value matches the source paper / re-derives consistently on both tracks
Confirmed w/ caveat	3	per-DN laterality bias not separable from the permutation null at small sample (DNa04, DNp26, DNng32)
Refuted	0	—

4 Discussion

The source paper traced a figure-ground circuit from a retinotopic sensory representation to wing-steering descending neurons and reported, in its supplementary tables, each figure-driven DN’s wing-steering output split by wing. This work independently confirms that motor-side result and renders it at per-edge resolution. Two contributions stand together; we are careful not to overstate either.

Scientifically, we provide an independent reproduction plus a per-edge map. The source paper’s S14 (per-DN ipsi/contra fractions) and S15 (DNp26 muscle targets) are reproduced exactly from the connectome—an external check on the motor-side claims—and we additionally emit the underlying per-edge dataset (488 DN $\rightarrow$ MN edges, 139 motor neurons, 69 muscles, both-wing coverage) that those summary tables aggregate. We do *not* claim to have discovered bilateral structure absent from the source paper: S14 already reports the wing split for these DNs. What is new here is the verification and the edge-level resolution—the specific motor neurons and muscles, per wing, behind each published fraction. With that map in hand, the behavioral logic is concrete: the figure-driven DN population converts a single retinotopic figure signal into a structured set of bilateral (DNbe001), ipsilateral (DNa04), and contralateral single-wing (DNp26, DNng32, DNge094) steering commands onto identified wing muscles—the anatomical form required to turn toward or away from a detected object.

Methodologically, the result was produced by a closed-loop agentic system whose defining feature is that it cannot certify a result it has not verified: the gate fails any round with a refuted claim or incomplete coverage, the supervising model is structurally barred from altering verdicts, and the per-edge dataset is emitted only after the source paper’s summary values are reproduced. The deterministic pipeline guarantees progression; the language-model layer supervises, triages, and reports. This pairing—verifiable connectome tracing under autonomous, steerable supervision—offers a template for independently checking and edge-resolving other connectomic circuit claims at scale, with an anchor-then-resolve discipline that keeps the output faithful to its source.

Data and reproducibility. All claims derive from public connectomic resources: FlyWire FAFB (materialization 783) for the brain side and the male whole-CNS connectome (MCNS v1.0) for the motor side. The certified dataset (`muscular_projection_dataset.json`), the per-claim verdict ledger (`verification_results.json`), the bilateral split, motor-target, and muscle-assignment tables, and the agentic tracing system itself are committed under 5 - Muscular Projection/ in the project repository.